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Lesson Notes

MODULE 7 | LESSON 3

INFORMATION AND DISTANCE

Reading Time	30 minutes
Prior Knowledge	Traditional portfolio theory and utility theory.
Keywords	behavioral finance, bounded rationality, satisfice, prospect theory, cognitive biases, emotional biases.

In the last lesson, we performed several modifications on the correlation matrix to improve its information content. In this lesson, we take correlation as a point of departure as we explore other ways to quantify the relationship between variables.

1. Correlation and Distance

In the last lesson, we continued to learn techniques to improve the covariance matrix. Specifically, we modified the correlation matrix to reduce error and improve its information content in order to construct better portfolios. Correlation and the correlation matrix have been our focus because they do contain crucial information required by mean-variance optimization (MVO), which we continued working with despite its disadvantages and limitations. We are also aware from previous lessons of the limitations of correlation itself. For example, Pearson correlation, which is the correlation implied by our use of the covariance matrix in MVO, captures linear relationships between variables. So when we restrict ourselves to traditional MVO, we are implicitly deciding to ignore non-linear information that might help us construct a better performing (higher return, lower risk) portfolio. This is reason enough to explore ways to capture non-linear relationships as well as portfolio construction techniques that take advantage of all available information about relationships between variables, whether these relationships are linear or not.

But before we turn our attention to information theory for its insights into the quantification of information about the relationships between variables and the structure of data, there is another important drawback to the use of correlation in its standard form: Say we want to organize a set of assets into groups based on their similarity, so that for example we can better control the level of concentration or diversification in the portfolio we construct. A natural way to approach this is to cluster the assets. As an example of such a strategy, recall the risk parity asset allocation strategy

we explored in Module 5 (“A Framework to Perform Asset Allocation Based on Partitional Clustering,” Duarte and De Castro). As a quick review, recall from that reading:

“*Clustering* is a term used to designate numerical methods of multivariate data analysis for the purpose of discovering groups of data. It can be defined as the organization of a set of objects into groups based on their similarity. Grouping is the process of partitioning a dataset into subsets so that the objects in each group share common characteristics, usually proximity, based on some measure of similarity or distance” (Duarte 11077).

We have of course our layperson’s understanding of the word “distance,” but a measure must technically satisfy three conditions before it can rightfully be called a distance in the sense of a metric space as required here:

1. **It must be non-negative.** This makes sense in our common sense understanding of the concept of distance: If you and I are at the same location, there is zero distance between us. There can of course be more distance between us if we move apart, but there cannot be less distance between us than zero distance, i.e., there cannot be negative distance. As a corollary to this requirement, there is actually another related condition, namely: There is zero distance between me and me. Conversely, if there is *zero distance* between us, such that you and I are in *exactly* the same location and taking up *exactly* the same space, then we are actually the same person. This may sound silly, but it is also logical and true.
2. **It must be symmetrical.** This also makes sense in our common sense understanding of the concept: It means that the distance is the same from my house to your house as from your house to my house.
3. **It must satisfy the triangle inequality.** This one also makes sense using our lay understanding of distance: If Ade lives 1 km from Ife and Ife lives 2 km from Ola, then Ade’s house can be no more than 3 km from Ola’s house.

We can see right away that correlation fails the first condition, and therefore, correlation is not a distance since correlation can be negative and distance cannot. As you will see in the readings (e.g., Marti et al. 2), there is a standard and straightforward way to overcome the negativity problem and convert correlation into a proper distance metric. By the way, this same technique was presented in the Duarte paper (11079) before using the *k*-medoids algorithm to cluster the assets based on a correlation distance.

With that said, the transformed correlation distance is still a product of the Pearson correlation, which itself only captures linear relationships between variables. As discussed above, we want to take advantage of all possible information in our portfolio construction, not just information about linear relationships. Other problems with Pearson correlation include that it is sensitive to outliers and it is possibly not appropriate for non-Gaussian distributions (Marti et al. 2). For these reasons, we look to other measures to help us make better financial decisions like asset allocation and portfolio construction.

2. Entropy

As you will learn in the required reading by Zhou, many disparate studies make use of the concept of entropy, including thermodynamics (where it originated), classical physics, quantum mechanics, and even sociology. For us, the most relevant understanding is related to probability and information

theory: Entropy measures the dispersion or uncertainty of a random variable, not unlike the concept of variance with which we are very familiar (but different). Since we generally want to minimize uncertainty—in finance, in machine learning, and in life—entropy can be quite useful. You will read in Zhou about how entropy has been used in portfolio selection in various consequential ways:

1. as a measure of risk (Zhou 4916)
2. as a measure of capital increment (Zhou 4919)
3. as a measure of portfolio diversification (Zhou 4920)

Entropy has also been used to price options (with fewer restrictions than Black-Scholes) and other derivatives, as well as model risk, utility, contagion, and commodity markets (Zhou 4922–6).

There is a daunting number of variations of entropy presented in Zhou's review. Try to appreciate the significance of each of them as you read, but ensure you know the meanings of **marginal entropy**, **joint entropy** and **conditional entropy**, as well as the definition and formula for the **Kullback-Leibler Divergence** (KLD), referred to in the reading as Kullback Cross Entropy (Zhou 2912). In addition to the elaboration on these metrics in the readings, see the Python code below for both practical calculations as well as intuition about their relationships to each other (and to mutual information, discussed in the next section). In addition, know and understand the following two principles, defined here for your convenience but expanded on in the readings:

The **Minimum Cross-Entropy Principle** is used to "identify the distribution P that satisfies a set of constraints and is closest to a target distribution Q , where the 'closeness' is measured by the Kullback–Leibler divergence" (Abbas 2).

Jayne's **Maximum Entropy Principle** states that "the probability distribution which best represents the current state of knowledge is the one with the largest entropy" (Cheng 2). This principle helps us specify the best approximation to an unknown probability distribution: The approximating distribution is the most uniform, makes no subjective assumptions about unknown conditions, and has the least risk of leading to an incorrect prediction (Cheng 3).

3. Mutual Information

While entropy may have some conceptual similarity to variance, **Mutual Information** (MI) is in some ways similar to correlation—but again, they *are* different. MI is a concept that is based on entropy: Specifically, MI is the reduction in entropy of one variable when another variable is known. An example would be helpful here: Suppose X is the result of rolling a six-sided dice, and Y indicates whether the result is odd or even. As Li notes, "Clearly, the value of Y tells us something about the value of X and vice versa. That is, these variables share mutual information" (3). You can also see clearly that MI is distinct from (though not unrelated to!) correlation. The required reading by Libman includes a nice discussion of this relationship between MI and entropy, before using MI to investigate the information—and mutual information—contained in various layers of the order book (the buy/sell orders aggregated by price level at a stock exchange). By the same token, the required reading by Guo examines the similarities and differences between MI and correlation by using each to build two separate hierarchical networks for purposes of comparison. Constructing these networks involves a clustering technique, as we saw in Duarte's approach to risk parity.

4. Python Implementation

When working with entropy, KLD, and MI in Python, you will find that the `scipy` library makes these calculations trivial. As you can see in the code below from Moen's repo ("ch3_metrics.py"), joint entropy and conditional entropy include MI-based terms because, as mentioned above, entropy and MI are closely related, conceptually and mathematically.

Due to this heavy reliance on `scipy`'s `mutual_info_score()` method, the documentation for it is also a required reading.

```
hX = ss.entropy(np.histogram(x, bins)[0]) #marginal entropy

hY = ss.entropy(np.histogram(y, bins)[0]) #marginal entropy

iXY = mutual_info_score(None, None, contingency=cXY)

hXY = hX+hY - iXY #joint entropy

hX_Y = hXY-hY #conditional entropy

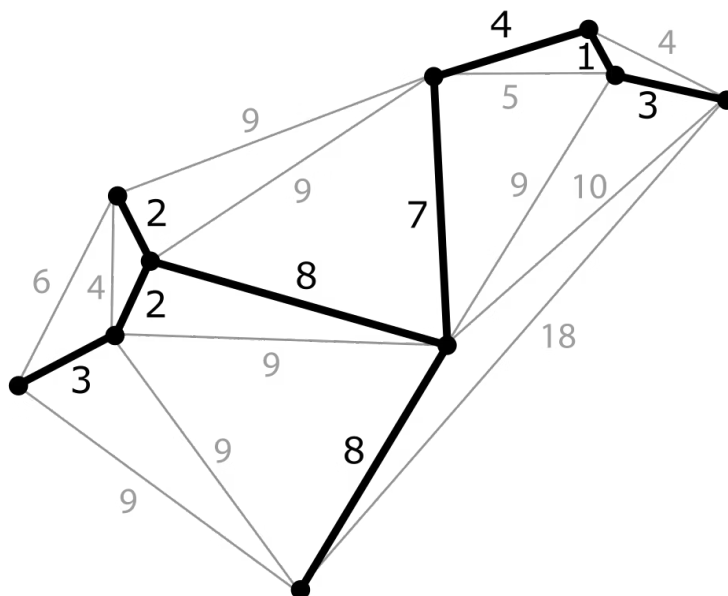
hY_X = hXY-hX #contitutional entropy
```

Note that if we had passed *two* probability arrays to the `entropy()` function, then it would compute the KLD, but here, we only pass one probability array, so the output is the entropy, as desired.

5. Connecting the Stocks

We are now familiar with a few of the most useful metrics for describing informational relationships between objects (e.g., stocks or other assets) or other financial phenomena (e.g., the layers of a stock exchange order book), as well as their applications in research related to financial decision making. In "A Review of Two Decades of Correlations, Hierarchies, Networks and Clustering in Financial Markets," Marti et al. contextualize the approaches we have seen in this lesson as well as the clustering in Duarte from Module 5. The article provides a useful gloss on "the relevant material from different fields, e.g. machine learning, information geometry, econophysics, statistical physics, econometrics, behavioral finance" (2). It also includes practical implementation details, such as how to constructing the minimum spanning tree (MST) using Kruskal's algorithm (about which there is a short and easy required reading [video](#)).

Figure 1: Example of a Minimum Spanning Tree (MST)



Source: "Minimum Spanning Tree." *Wikipedia*, Wikimedia Foundation, 30 Mar 2022.
https://en.wikipedia.org/wiki/Minimum_spanning_tree.

MST is the method used in Guo to build a network diagram for the Chinese stock market, and it was also the clustering approach used in the Duarte article from Module 5.

At least some of the methodological concerns mentioned should be familiar to you by now, e.g., the problems with Pearson correlation (Marti et al. 2). The following sections are especially rich with valuable insights:

- "Dynamics of Correlations, Hierarchies, Networks and Clustering" (Marti et al. 6)
- "Financial Applications" (Marti et al. 10)
- "Practical Fruits of Clusters, Networks, and Hierarchies" (Marti et al. 15)

6. Conclusion

We learned how to transform correlation into a distance metric, which can be used to improve the portfolio construction process, especially from the perspective of diversification. Due to correlation's weaknesses, we learned about measures from information theory to help us quantify relationships between variables in order to organize the variables into clusters and networks. In the next lesson, we continue to apply clustering techniques as we investigate the strengths and weaknesses of hierarchical risk parity.

References

- Abbas, Ali et al. "A Kullback–Leibler View of Maximum Entropy and Maximum Log-Probability Methods." *Entropy*, vol. 19, no. 5, 2017, pp. 1–14, <https://www.mdpi.com/1099-4300/19/5/232>.
- Cheng, Ge et al. "Predicting the Outcome of NBA Playoffs Based on the Maximum Entropy Principle." *Entropy*, vol. 18, no. 12, 2016, pp. 4909–4931, <https://www.mdpi.com/1099-4300/18/12/450>.
- Duarte, Flavio Gabriel, and Leandro Nunes de Castro. "A Framework to Perform Asset Allocation Based on Partitional Clustering." *IEEEExplore*, vol. 8, 12 Jun 2020.
<https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9115637>.

- Guo, Xue et al. "Development of Stock Correlation Networks using Mutual Information and Financial Big Data." *PLOS One*, 18 Apr 2018, <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0195941>.
- Libman, Daniel et al. "Mutual Information between Order Book Layers." *Entropy*, vol. 24, no. 3, 2022, <https://www.mdpi.com/1099-4300/24/3/343>.
- Marti, Gautier et al. "A Review of Two Decades of Correlations, Hierarchies, Networks and Clustering in Financial Markets." *Progress in Information Geometry*, 2017, pp. 245–274, <https://arxiv.org/abs/1703.00485>.
- OCLPhase2. "Kruskal's ex 1." _YouTube, <https://www.youtube.com/watch?v=gaXM0HNErc4>.
- Zhou, Rongxi. "Applications of Entropy in Finance: A Review." *Entropy*, vol. 15, no. 11, 2013, pp. 4909–4931., <https://www.mdpi.com/1099-4300/15/11/4909>.

